

Question:

A group of scientists is conducting an experiment on the human cells involved in the immune response. They genetically modify B cells so they do not express the cluster of differentiation 21 (CD21) on their cell surfaces. The pathogenesis of which of the following organisms would most likely be affected by this genetic modification?

Answer Choices:

- A. Epstein-Barr virus (EBV)
- B. Measles virus
- C. Human immunodeficiency virus (HIV)
- D. Human papillomavirus

Reasoning + Answer:

<think>

- CD 2 1 is a receptor on B cells that helps in the activation of B cells by the interaction with CD 1 9 and CD 8 1 . It also plays a role in B cell signaling and antibody production .

- EB V and measles both have the ability to infect B cells and use them for their replication and spread . Specifically , EB V uses CD 2 1 as a receptor to enter B cells , which are crucial for its replication cycle .

- HIV primarily targets and destroys CD 4 + T cells , not B cells .

- Human pap ill om avirus (HPV) infects epith elial cells and does not have a dependency on CD 2 1 for its infection cycle .

Given these points , the genetic modification of B cells not to express CD 2 1 would most affect the pathogenesis of Epstein -B arr virus (EB V) , as EB V uses CD 2 1 as a receptor for infecting B cells .

</think>

<answer>A . Epstein -B arr virus (EB V)</answer>